Lecture Transcript by Slide

Week 12 (posted for Weeks 12 and 13) · Appendix E: Pacemakers · 29 min
Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

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1 Appendix E: Pacemakers (title slide)

0:00:00 – 0:01:27



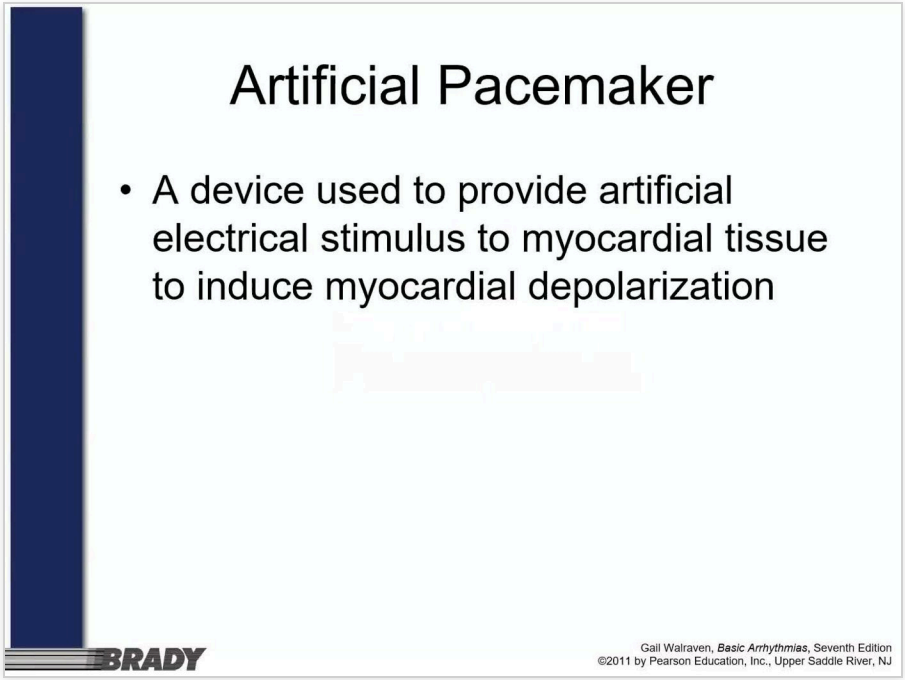
0:00:11 Michael V Hyland: Alright. So this is gonna be the last of our new content. Michael V Hyland: So the last set that we're going to talk about is pace rhythms.

0:00:26 Michael V Hyland: Now, there's a lot of information in this, Powerpoint. I'm not going to test you on, so as we go through this, I will let you know what you need to know and what you don't need to know. Michael V Hyland: Basically, what I'm going to want you to be able to do is identify whether the rhythm is paced or not, and tell me whether it's paste. Michael V Hyland: atrial paste, ventricular paste, or a B paste. and it's very easy to determine whether it's a paste v. Paste or AV paste Michael V Hyland: for those of you that did, ekg. Clinical. Were you looking at any of the telemetry monitors while you were there?

0:01:05 Michael V Hyland: And you saw that the patients that were paced, they color coded the rhythms purple on the screen. And what were you? How do we know those rhythms were paced Michael V Hyland: by a pacer spike. So there's a vertical spike right on the screen, right that showed that beat or that rhythm was paste. Michael V Hyland: So let's quickly go through some of this information here. So artificial pacemaker.

2 Artificial Pacemaker

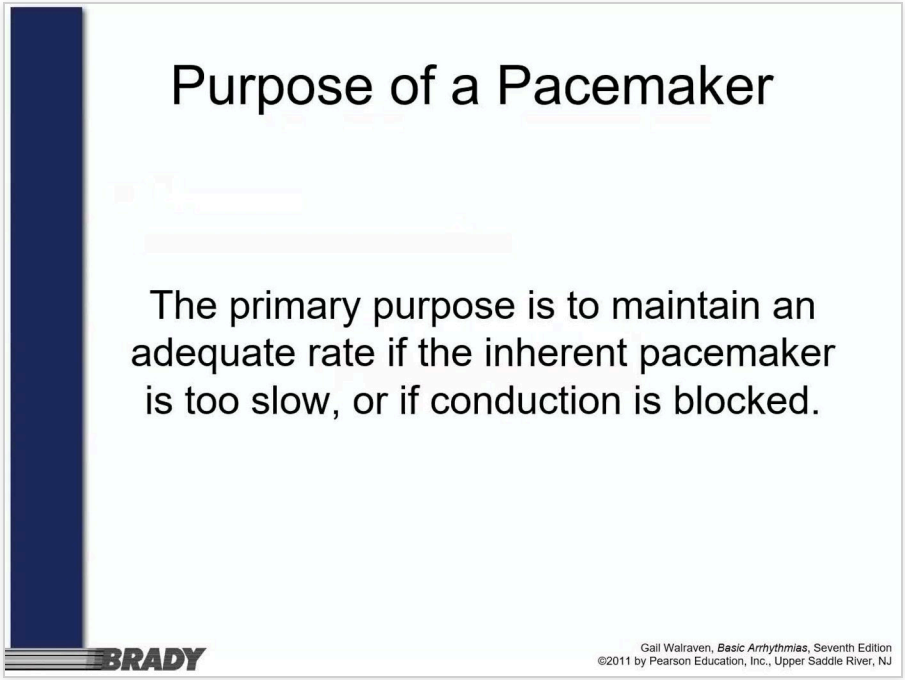
0:01:27 – 0:01:44



0:01:31 Michael V Hyland: it's typically a device used to provide an artificial stimulus to the patient's Michael V Hyland: harp to cause depolarization. Well, when would we need this?

3 Purpose of a Pacemaker

0:01:44 – 0:02:45



0:01:45 Michael V Hyland: We would need this if the patient's inherent pacemaker, IE. The sa node right is actually firing too slow for the to sustain that patient's life. Michael V Hyland: or if there is Michael V Hyland: complete heart block and the pacer. that whatever is pacing the patient's lower 2 chambers, typically either the junction or the ventricles generally is too slow, and we need a faster heart rate. Michael V Hyland: So those are the 2 reasons why why Michael V Hyland: a patient would have a permanent pacemaker placed the inherent pacemaker is too slow, or they've been diagnosed with third degree heart block.

0:02:32 Michael V Hyland: And again, I'm not gonna test you on Michael V Hyland: describing what a pacemaker is, or indications. Why, we would place it. This is Ekg interpretation. So I'm gonna be looking for you to identify pace rhythm. So I will discuss that when we get there. So how do we identify that a beat

4 Pacemaker Spikes

0:02:45 – 0:04:04

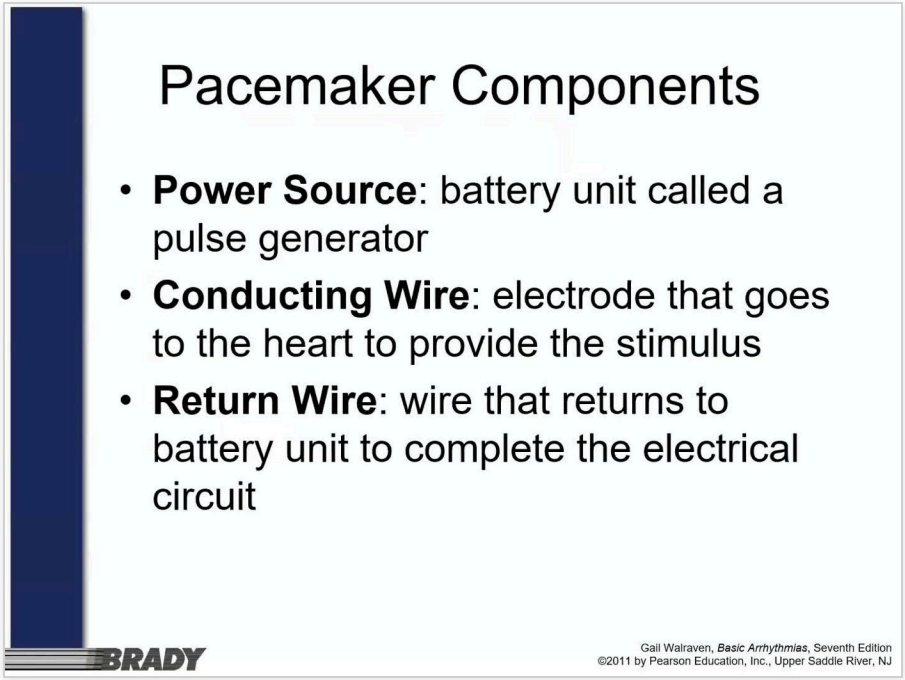


0:02:51 Michael V Hyland: is paced or rhythm is paste? Because when the pacemaker fires, it creates a vertical line Michael V Hyland: on the Ekg strip Michael V Hyland: that causes depolarization. Michael V Hyland: So if the patient is apaced. Michael V Hyland: we'll see a pace or spike where the P. Wave should be. Michael V Hyland: and then we'll see it, followed by a QRS. So what happened is the pacemaker sent that signal Michael V Hyland: to the atrium muscle cells. The atrium muscle cells contracted, and then it sent that signal down through the ab node, the bundle of his, the right and left bundle branches, and then we see Qrs.

0:03:37 Michael V Hyland: If the patient has a ventricular pace beat. Michael V Hyland: We're not going to see a P. Wave. Michael V Hyland: We're going to see a spike a vertical line at the very beginning of the Qrs. Michael V Hyland: it's that simple. If somebody is avaced, we're going to see a spike where the p-wave should be, and a spike at the beginning of the Qrs. Michael V Hyland: It really is that simple? Okay?

5 Pacemaker Components

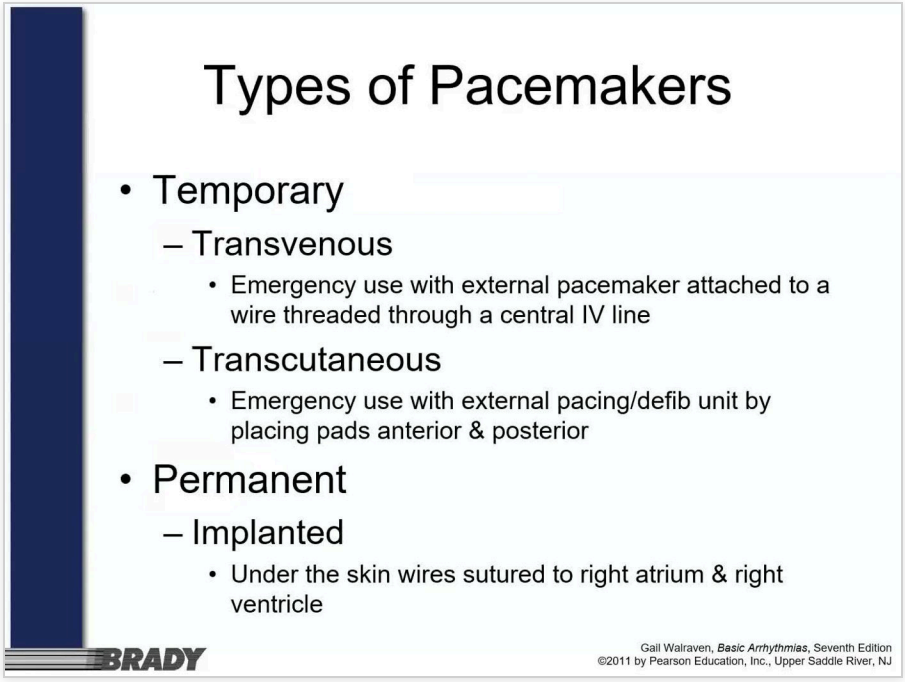
0:04:04 – 0:04:25



0:04:06 Michael V Hyland: So none of this stuff is going to be testworthy information. So Michael V Hyland: it's not a pacemaker course. So I'm not really gonna cover any of this right? This is just talking about. You need a power source, a wire going from the pacer Michael V Hyland: to the heart muscle, and then a return wire. There are typically 3 types of pacemakers that we use.

6 Types of Pacemakers

0:04:25 – 0:05:48

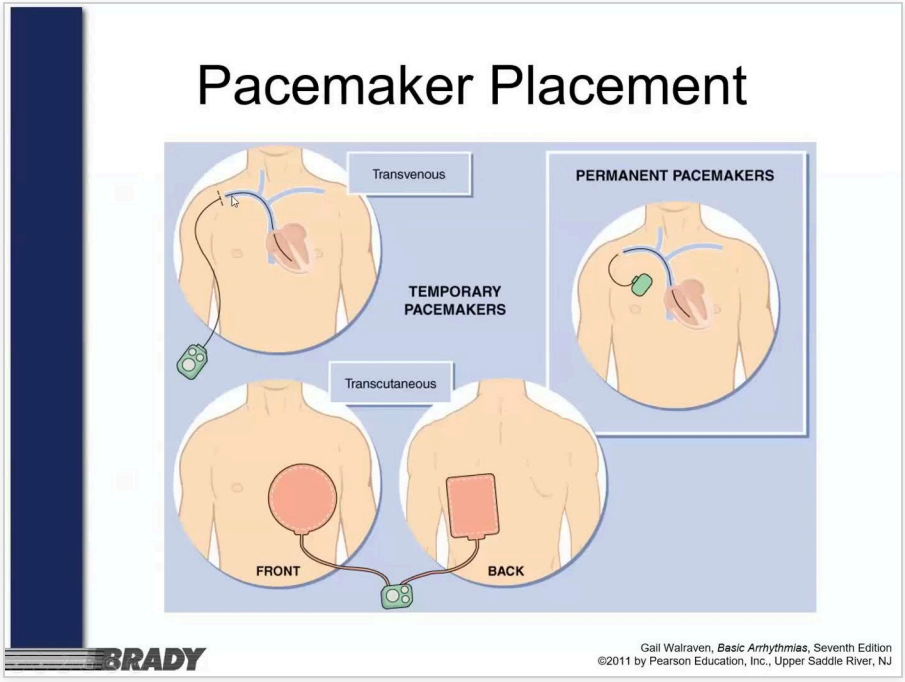


0:04:33 Michael V Hyland: We can actually put a central Venus IV. Capital in the patient. and we can thread a wire through that central IV line Michael V Hyland: hook that wire up to a pacemaker we call that transvenis pacing. Michael V Hyland: So we have a wire going through a catheter, that catheter is in a central line that goes to the heart. The wire leaves the catheter and goes inside the heart, and so it touches the heart muscle to to deliver that Michael V Hyland: stimulus for depolarization Michael V Hyland: transcutaneous. So we can put a big pad on the patient's chest and the back, and then we can deliver that stimulus to the pad, and that electrical stimulus goes across the skin.

0:05:24 Michael V Hyland: Cutaneous means skin. So that's why we call it transcutaneous across the skin or across the vein. Michael V Hyland: and then they can surgically implant a pacemaker underneath the skin that's usually up on the right shoulder looks like a little deck of cards underneath the patient's skin, and then they hook wires directly to the heart muscle itself.

7 Pacemaker Placement

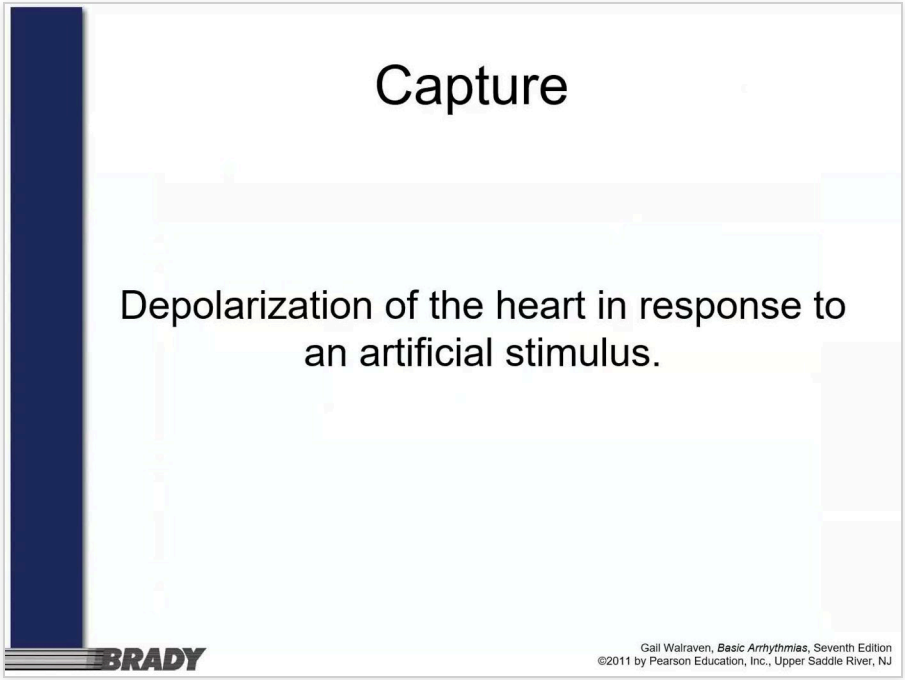
0:05:48 – 0:06:30



0:05:48 Michael V Hyland: and this is kind of what it looks like. So we place an ib. Michael V Hyland: Line into this vein, this big vein which runs underneath the patient's collarbone. and that cap that goes all the way to the junction of the superior vena cava and the right atrium. Michael V Hyland: and then we thread the wire through that catheter, and it goes down and touches the wall of the right ventricle. So that's transvenus transcutaneous. Michael V Hyland: Again we put in the patch on the front and back, and we pace through the skin, and then a permanent one. Here is the pacemaker box, and the wire is connected to the patient's heart again. Not information. I'm gonna test you on

8 Capture

0:06:30 – 0:07:48

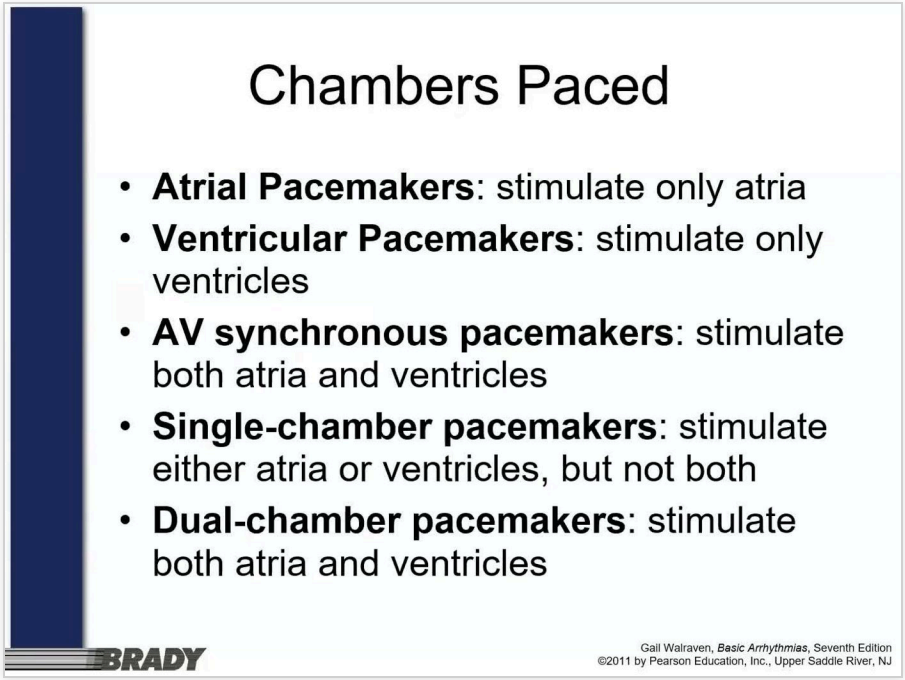


0:06:33 Michael V Hyland: capture Michael V Hyland: capture means. And again, I'm not going to test you on this. But what capture means is when we send a signal from the pacemaker to the heart. Does it cause it to depolarize, meaning that impulse captured the heart and caused it to contract. Michael V Hyland: So all capture means is, did it make the heart depolarize? So when we do a pacemaker. We're setting up a pacemaker. We can set the rate at which the number of impulses gets sent to the heart muscle, and then we set the strength of the impulse. Generally we start off at a low level.

0:07:12 Michael V Hyland: right? Michael V Hyland: And it's it's kind of like a 10 unit. You guys know what a 10 unit is right. It delivers electrical impulses to your muscle right? When you're trying to heal, or when you have muscle spasms, you start off at a low level, and then you keep turning it up. Well, here we keep turning it up until we see the polarization Michael V Hyland: cause. If we see a spike on the on the Ekg strip, and there's not a waveform following it. Then that level of impulse did not cause the heart to depolarize.

9 Chambers Paced

0:07:48 – 0:08:06



0:07:50 Michael V Hyland: I'm going to skip this whole part, if you want to learn about this. That's awesome. But these are all of the different types of Michael V Hyland: ways that a patient's heart can be paced.

10 Smart Pacemaker

0:08:06 – 0:09:15

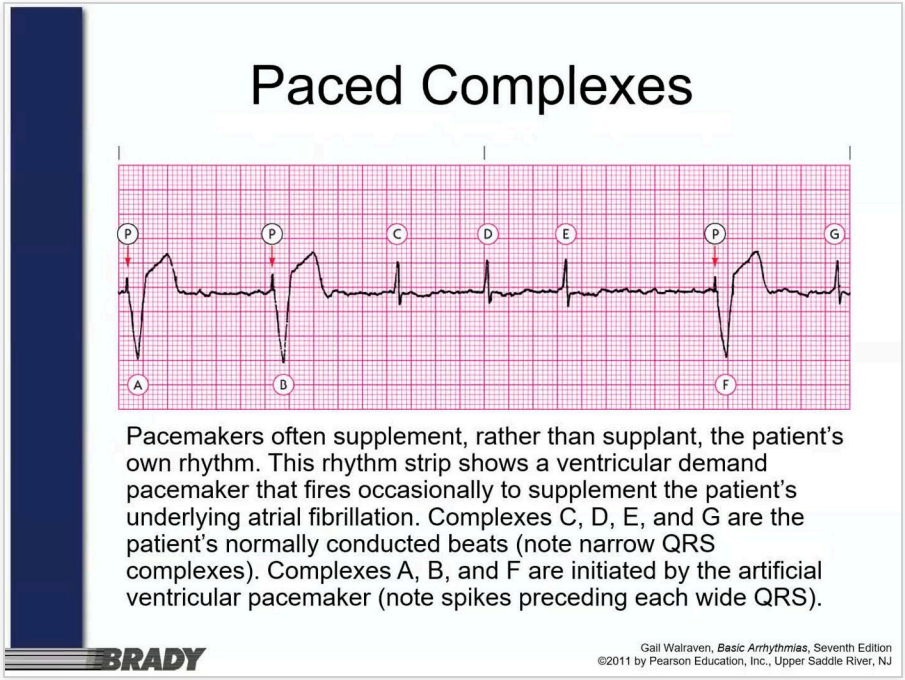


0:08:07 Michael V Hyland: and the newer pacemakers now are like your smartphone. Michael V Hyland: So it. And now it allows these newer pacemakers to sense intrinsic activity. Michael V Hyland: So some people might have some sinus node activity. Michael V Hyland: Maybe it's only 40 50 beats per minute. Right? So when the pacemaker sends the sinus, the essay node fires the patient's essay node fires, the pacemaker holds back and doesn't do anything, but if a certain amount of time elapses, then it sends an impulse Michael V Hyland: so it can sense intrinsic activity and pace only when needed.

0:08:51 Michael V Hyland: And these are the different responses. Again, I'm not going to test you guys on this Michael V Hyland: paste sensed response. And they do this three-letter pacemaker code. So I'm just gonna skip through this because I know you guys wanna just get to the Michael V Hyland: meat of what you need to know. All right. So I'm gonna focus on this rhythm just for a second.

11 Paced Complexes

0:09:15 – 0:12:27



0:09:30 Michael V Hyland: So you can see that each one of these beats is letters. The first one is A, BCDE, Michael V Hyland: FG, Michael V Hyland: so A, B and C are the beats that have been paste. Michael V Hyland: So if you look where the P is, there's a spike right here, so that spike Michael V Hyland: now produces a Qrs after it. And this is a ventricular paste beat. I know it's a ventricular paste beat because it looks like a PVC. Michael V Hyland: So when the wire is touching the ventricle, and we're only pacing the ventricles, we'll not have a p-wave in front of it.

0:10:16 Michael V Hyland: The the electrical impulse has to spread cell to cell. Michael V Hyland: so that impulse spreads slower. Michael V Hyland: So anytime we pace the ventricle, any beats that we see are going to be wide. Q. Rss, it's going to look just like a PVC.

0:10:36 Michael V Hyland: And R. And T. Will go in opposite directions. So anytime, you see a beat with a spike Michael V Hyland: with a wide Qrs. R. And T. Going in the opposite directions. You know that that is a ventricular paste beat.

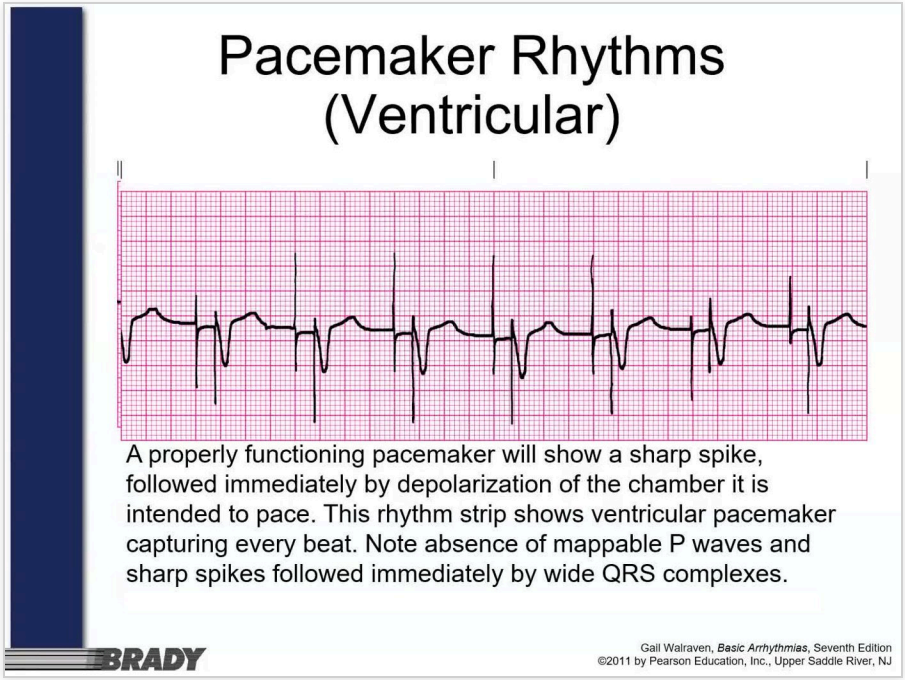
0:10:54 Michael V Hyland: and so you can see. A, B and F are all ventricular paste beats Michael V Hyland: all of the other ones. CDE and G are from the patient's intrinsic activity. There's no pacer spike there.

0:11:11 Michael V Hyland: Anybody want to take a guess? The underlying rhythm? Just look at CD. And E. You see any identifiable pea waste. Michael V Hyland: Nope, don't see any. What do you see? F waves? Lot of F waves. There are those CD and E. Space evenly, or you know, by the eyeball, I think I think they're going to be different. Right? Michael V Hyland: We counted them out (510) 151-6172 halves make 18, 5, 1011, 14, 2 2 halves make 15. So one is 18, one is 15, so they're not evenly spaced. What rhythm is irregular, completely irregular with f waves Michael V Hyland: hey? Fit so the underlying rhythm. Here is a fifth Michael V Hyland: with 3 ventricular paste beets. Now, I'm not going to do that to you before you start panicking. I'm not going to do that to you Michael V Hyland: when I when I get to the rhythms that I'm going to give you on the homework and on the test. In 2 weeks

0:12:17 Michael V Hyland: the entire rhythm will be paced, not some beats and not other beats. Michael V Hyland: You're welcome.

12 Pacemaker Rhythms (Ventricular)

0:12:27 – 0:18:50



0:12:27 Michael V Hyland: Alright. So I'm gonna do this one. We're gonna gather our information like we normally would. Michael V Hyland: As I'm going along. If you have questions, you need to stop me and ask your question. Michael V Hyland: there are no stupid questions.

0:12:48 Michael V Hyland: Okay, so just to orient you on this when we're looking at this right? I can't really tell what's happening with this first B, Michael V Hyland: this looks like the end of the Qrs with the T-way. But other than that, I really don't know what's going on there. Right. So now, as I go to the next beat, I see this big vertical line. That's a pacer spike. Michael V Hyland: right? And then I see another vertical line that's a pacer spike, and then I see a Qrs. Followed by a T wave. Michael V Hyland: and then that pattern repeats itself over and over again. Right? So our waves are down. T. Waves are up right? So I'm going to gather my information.

0:13:31 Michael V Hyland: 5, 1015, Michael V Hyland: 19, either 19 and a half or 20, Michael V Hyland: 5, 1015, Michael V Hyland: 19. I think that's 2 halves. I'll call it 20. So I'm going to say this spacing is 20,

0:13:55 Michael V Hyland: so I'm going to go 1,500, divided by 20, because my R to R is regular Michael V Hyland: right. That gives me a heart rate of what?

0:14:07 Michael V Hyland: 75. Michael V Hyland: So the rhythm is regular. Michael V Hyland: the heart rate is 75 pea waste. Do you see any P waves on this strip.

0:14:23 Michael V Hyland: I don't. But what I do see Michael V Hyland: is the pacer spike where the P. Wave should be. Everybody agree on that. So there's A. P. Wave where the pacer spikes should be. So what can I put for my P wave

0:14:40 Michael V Hyland: paste? Okay. PACE. D. These these P. Waves are paste. Michael V Hyland: Now, technically, I can measure the Pr interval here

0:14:55 Michael V Hyland: because I'm going to go from the Pacer spike to the beginning of the Qrs. Michael V Hyland: Or in this case from the pacer spike to the pacer spike.

0:15:06 Michael V Hyland: So what do I got? Looks like I got 4, one. Michael V Hyland: 2, 3, 4, Michael V Hyland: and that's probably pretty consistent. Who's confused by that.

0:15:20 Michael V Hyland: Alright. So let's look at this. Michael V Hyland: Find a pacer's fight. This one here starts right on that line. Michael V Hyland: and all I'm gonna do is count the number of small boxes from there to the beginning of the Urs. Michael V Hyland: 1, 2, 3, 4.

0:15:41 Michael V Hyland: This one is a little bit to the left of the line Michael V Hyland: as the Urs starts a little bit left in the line. So I'm like, half half halfway to halfway. Michael V Hyland: 1, 2, 3, 4, Michael V Hyland: everybody got that. Michael V Hyland: So this. Yeah. Ri is 4 small boxes point 1 6 s.

0:16:05 Michael V Hyland: What are the lines down there? Those are the pacer spikes. So they're going up as well as down. Michael V Hyland: This is Michael V Hyland: a pacer spike for atrial contraction. This is a pacer spike for ventricular contraction. Michael V Hyland: Don't let the lines confuse you all. It is is a patience. Michael V Hyland: So you're still measuring from the beginning of the the atrial pacer's bike to the beginning of the Qrs. Michael V Hyland: Which happens despite despite in this particular case.

0:16:45 Michael V Hyland: Now we're going to measure the width of our Qrs. Michael V Hyland: I know this Qrs is going to be. Why? Because the ventricular beats are paste right? R and T in opposite directions.

0:17:02 Michael V Hyland: So we're gonna go right from here Michael V Hyland: to here. Michael V Hyland: 1, 2, 3, and it might even be slightly bigger. Michael V Hyland: Then 3 boxes right for this. We're gonna go right from here. It's beginning in the Qrs. Right there all the way over to them. Michael V Hyland: 1, 2, 3, Michael V Hyland: 1, 2 Michael V Hyland: great. Everybody got that? Sure.

0:17:41 Michael V Hyland: So this is the beginning of the Qrs right here. right? Michael V Hyland: And then this is the end of it. 1, 2, 3, the next one right there it ends there. 1, 2, 3, same measurement. Right

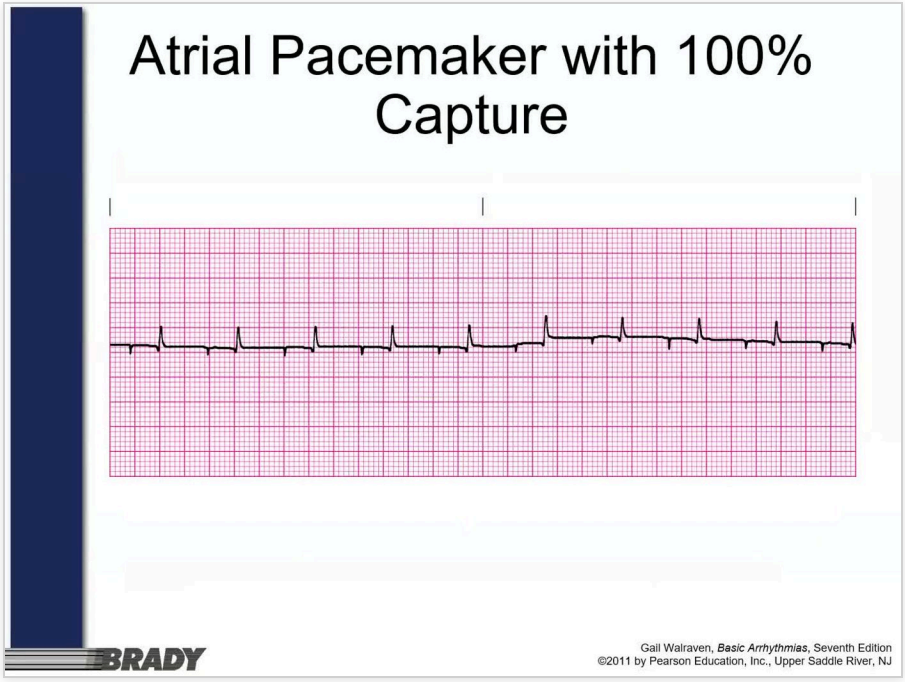
0:17:59 Michael V Hyland: now. How do we interpret this? I know that the atria being paced because I see atrial pacer spikes. I know that the ventricles are being paced because I see ventricular pacer spikes. Michael V Hyland: so all I would expect you to write here would be a capital letter A, followed by a capital letter V, Michael V Hyland: followed by the word paste. This is an A. V paste rhythm.

0:18:26 Michael V Hyland: Everybody comfortable with that Michael V Hyland: you have to be Michael V Hyland: because I will put an A B paste rhythm Michael V Hyland: on the test. In 2 weeks there will be an avast rhythm on the test.

0:18:44 Michael V Hyland: and probably on the final example. Not probably definitely. All right. Let's look at this river.

13 Atrial Pacemaker with 100% Capture

0:18:50 – 0:22:01



0:18:55 Michael V Hyland: So let's gather our information. Our waves are up now. so let me let me Orient you? What is this line right here?

0:19:09 Michael V Hyland: That's a pacer spike. and that's a pacer spike where the atria, or where the p-wave should be right. And then I have kind of a long Pr interval Michael V Hyland: to a negative Q wave, followed by a positive R wave. Michael V Hyland: and then I don't really have an S wave again. We haven't been paying too much attention to that right. and there's no pacer spike near the Qrs. Any guesses to how we would interpret this rhythm

0:19:38 Michael V Hyland: we have atrial pacer spikes with a narrow QRS. Michael V Hyland: Well, if what did we do last time? Wherever there was a pacemaker Michael V Hyland: spike we use that in the interpretation. Michael V Hyland: So this is just going to be Michael V Hyland: a paste or atrial paste, right? Michael V Hyland: Because we only have pacer spikes Michael V Hyland: for the P waves, or where the atria is right. there is no t-wave here. Michael V Hyland: there's no t-wave.

0:20:15 Michael V Hyland: So again we would count our wave to our wave. This rhythm is going to be regular. I don't know how many small boxes, 5, 10, Michael V Hyland: 1415, and 2 halves make 16. So probably 16. So this is going to be a heart rate of like 97, 98, Michael V Hyland: 1,500, divided by 16

0:20:39 Michael V Hyland: 94. Michael V Hyland: The rhythm's regular, with a heart rate at 94 Michael V Hyland: p. Waves. What are we writing for the P. Waves Michael V Hyland: paste? There's a spike there. So we're writing paste. Michael V Hyland: We can measure the Pri. We go from the pacer spike to the beginning of the Qrs.

0:21:03 Michael V Hyland: And it looks like it's going to be 5 small boxes Michael V Hyland: beginning of the pacer spike to the beginning of the of the waveform actually might be 5 and a half. If I look at this starts right on that line right there Michael V Hyland: and then the Qrs starts right there halfway in between. So that would be 5 and a half.

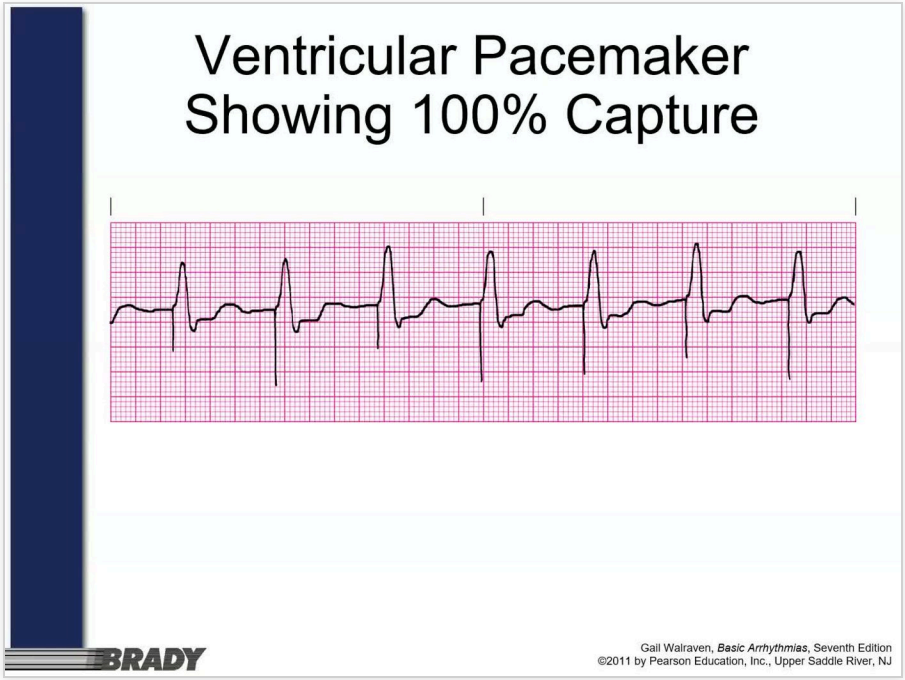
0:21:27 Michael V Hyland: So I think this, I think that this Pri is 5 and a half small boxes that would give me point 2 2

0:21:36 Michael V Hyland: QRS. Going to be pretty narrow. Michael V Hyland: one and a half. Michael V Hyland: one and a half point 0 6, and my interpretation would be a paste

0:21:55 Michael V Hyland: questions about that Michael V Hyland: alright. And then this is going to be the last one. So there's only going to be 3 rhythms out of this pacing section. a paste be paste or a B paste.

14 Ventricular Pacemaker Showing 100% Capture

0:22:01 – 0:25:06



0:22:10 Michael V Hyland: So I kind of look at, you know, the last 2 chapters with the ventricular and the heart blocks were really tough. but now I'm kind of giving you a gift they're pretty easy to identify. Michael V Hyland: Oh, my goodness! There's a vertical line. What is that? That's a pacer spike. Okay, what chambers are being paste a paste be paste or both.

0:22:32 Michael V Hyland: So let's gather our information. You clearly see the pacer spikes here at the beginning of the Qrs. Right? And there's no p waves on this rhythm strip at all. So let me orient you. Michael V Hyland: This is T. Wave. Michael V Hyland: isoelectric line pacer, spike, QRS. Michael V Hyland: T. Wave. and then repeat itself. Pacer spike. Qrs. Michael V Hyland: okay. let's gather our information. Michael V Hyland: 5, 1015, 20, and a half. 5, 1015, 20, and a half, so I'm pretty satisfied. This is 20 and a half small boxes between our waves.

0:23:26 Michael V Hyland: our rate Michael V Hyland: 73. Michael V Hyland: He waves. Michael V Hyland: you see a pacer spike where the P. Wave should be.

0:23:43 Michael V Hyland: I don't. Michael V Hyland: I don't see any P. Waves, and I don't see a pacer spike. Oh, I see I see a pacer spike, but I see a ventricular pacer spike, not an atrial pacer spike. Michael V Hyland: so P. Waves none.

0:23:59 Michael V Hyland: Pr. I done. Michael V Hyland: Now. We got to measure the Qrs. This is a pretty wide QRS. Michael V Hyland: So we're going to include this little dip over here. This is so. The pacer spike starts it. Michael V Hyland: The end of this little dip ends it. 1, 2, 3, 4 and

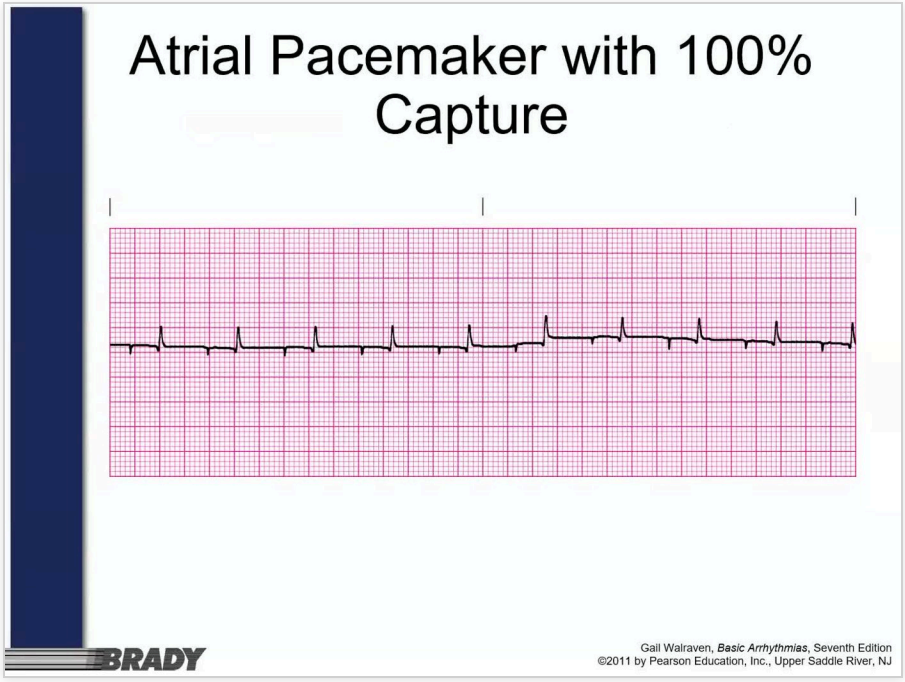
a half.

0:24:23 Michael V Hyland: 1, 2, 3, 4, 5, Michael V Hyland: one. 2, 3, 4, 5. So I'm going to believe this is 5 small boxes. So the width of this Q. Rs is point 2 0,

0:24:40 Michael V Hyland: and then my interpretation is going to be V paste Michael V Hyland: questions, comments. Yes. okay. So Michael V Hyland: books them comes like, and digital. They both come before mixed through our rest. Michael V Hyland: So how did you know, like Michael V Hyland: April? Look at where the patient spike is.

15 Atrial Pacemaker with 100% Capture (revisited)

0:25:06 – 0:26:13



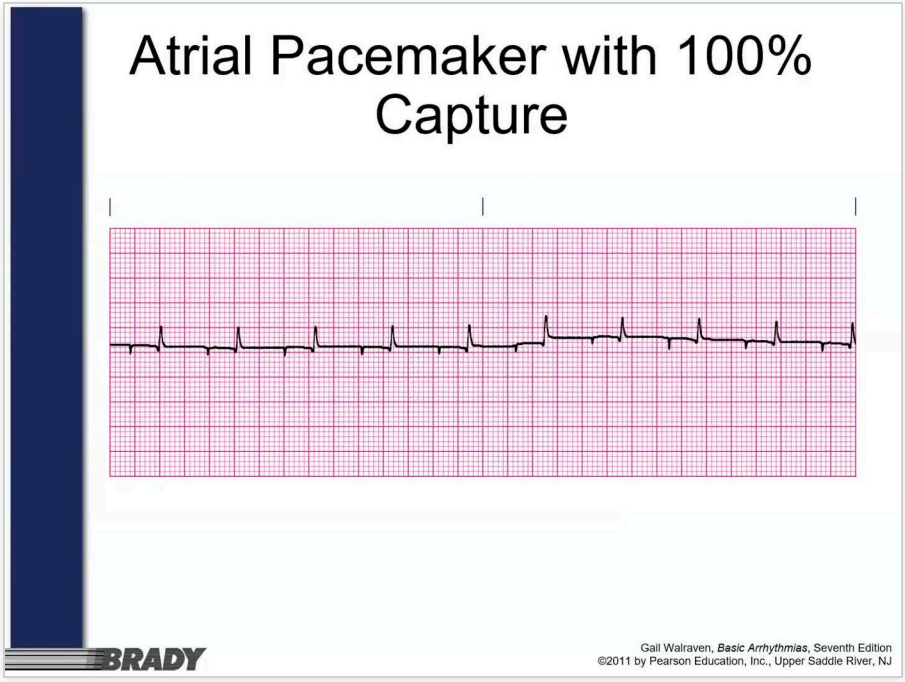
0:25:10 Michael V Hyland: There's you know that these under Q. Rss, you can identify that there's no and these were based. They would have a spike in the beginning of it.

0:25:22 Michael V Hyland: There's a spike right there where the p-wave should be. Michael V Hyland: So that's how I know that the atria is being paced. Michael V Hyland: But then once that atri is paste, that single fee has to be a no, the bundle of his. Michael V Hyland: and above the branches, and is now, because we pasted here now, the metrical dark attracting, based on the pacemaker that we set. But the spike appears at the P. Wave, not at the Qrs.

0:25:54 Michael V Hyland: Got it. Did that answer your question so like like this one. there are no pea waves, and the spike is at the beginning of the QRS. Michael V Hyland: So you're locating where you see the pacer spikes and tell me what chamber is being paced. And so

16 Atrial and Ventricular Pacemaker Strips (comparison)

0:26:13 – 0:26:56



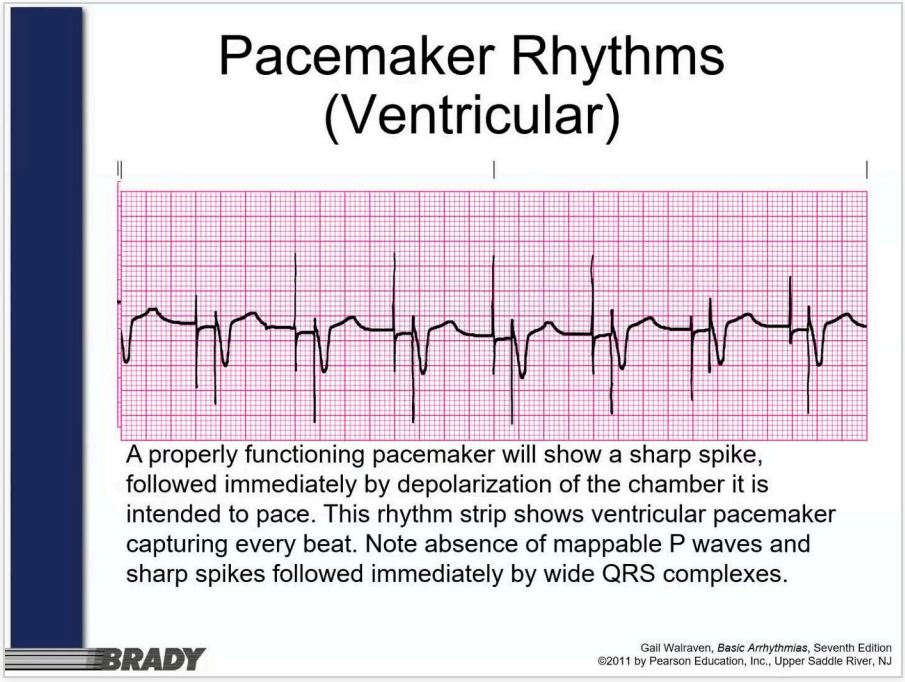
0:26:13 Michael V Hyland: when we go to a paste. And so you just have to remember some details. You see where the pace is. Spike is. You automatically know what chamber is being paste Michael V Hyland: right? And then it's easy. You're counting R to R for the regularity and the heart rate, because most of these pace rhythms, all the paste rhythms I give you are going to be regular. Michael V Hyland: They're all gonna be regular

0:26:37 Michael V Hyland: to get your heart rate, and then you see the pace of spiance. So if there's a pace of spike where the atri should be, then the P. Waves are paste.

0:26:47 Michael V Hyland: and then you can still measure your Pr. Interval. your the width of the Qrs, and then your interpretation.

17 Pacemaker Rhythms (Ventricular) (revisited)

0:26:56 – 0:27:32



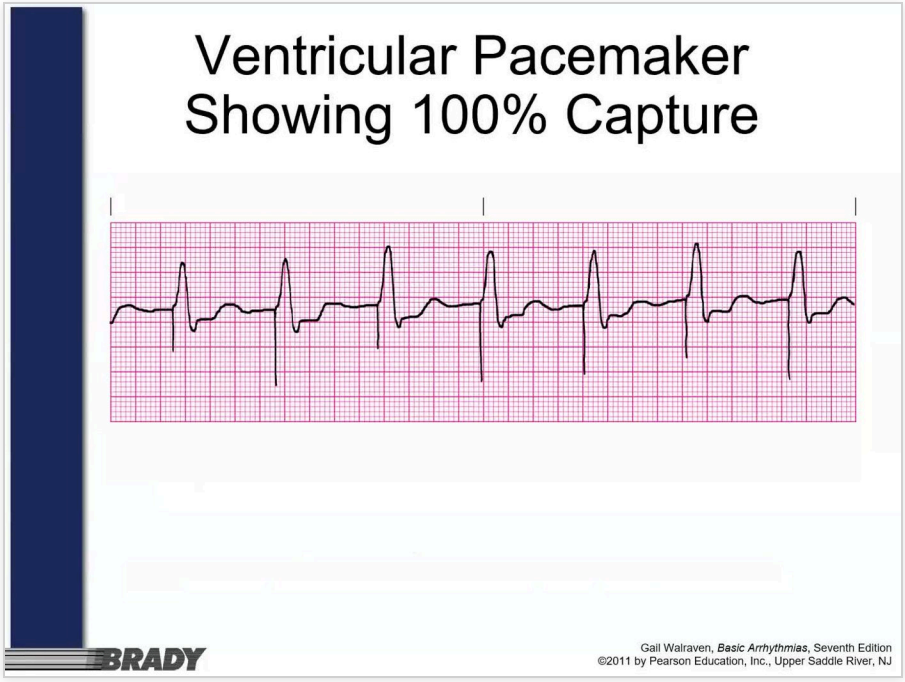
0:26:56 Michael V Hyland: If it's a B paste. You're gonna see 2 spikes for each beat. Michael V Hyland: Still, the same thing, our our regular.

0:27:07 Michael V Hyland: that's going to help me with my heart rate. I see a patient fight where the p-way should be. I know I'm a pacing. I see a pacers fight at the beginning of the ventricle. I know I'm B. Pacing.

0:27:20 Michael V Hyland: So my P. Waves paste, I can still measure my Pr interval, measure the width of my Qrs, and then my interpretation.

18 Ventricular Pacemaker Showing 100% Capture (revisited)

0:27:32 – 0:29:00



0:27:32 Michael V Hyland: The V. Pacing is the only time I'm not going to have P. Waves, and I'm not going to have a pri

0:27:43 Michael V Hyland: questions, comments Michael V Hyland: alright. So that's gonna conclude the pacemaker Michael V Hyland: rhythms. Michael V Hyland: I think, on the homework that I handed out today, which is going to be your last homework assignment for the semester. There's 2 or 3 paste rhythms on that homework assignment. Michael V Hyland: There's also heart blocks on that homework assignment. There is also Michael V Hyland: ventricular rhythms on that heart on that homework assignment. So next the next time we meet, you're gonna have an hourly exam on that hourly exam is gonna be ventricular rhythms, heart blocks and paste rhythms.

0:28:36 Michael V Hyland: Okay? Michael V Hyland: So the homework we have based ventricular and park logs. Okay?